

W1 WORD VECTORS

NLP pipeline: text → tokens → repr → model → decode/eval; always state task, unit, labels, cost.
Text units: char/subword/word/span/doc; smaller vocab helps OOV, longer hurts context cost.
BoW: sparse counts $x_i = c_i(t, d)$; loses order, strong baseline for topic. Use dict/sparse vec when $|V|$ large: store non-zero only.

TF-IDF: $t_{f,i,d} = 1 + \log_{10} c_i(t, d)$ if $c > 0$; $i d_{f,i} = \log_{10} (N/d_{f,i})$; $f_{i,d} = t_{f,i,d} \cdot i d_{f,i}$.
IDF intuition: high raw freq can be useless if term occurs in all docs; IDF boosts rare discriminative words.
One-hot: orthogonal, no similarity; dimension $|V|$; okay for lookup, bad semantic distance.
Distributional hyp: words in similar contexts have similar meanings; static vectors compress co-occurrence.
Word2Vec: CBOW predicts center from ctx; Skip-gram predicts ctx from center; neg sampling converts softmax to binary discrimination.
 $\cos(u, v) = u^T v / (\|u\| \|v\|)$; analogies rely on linear offsets, not guaranteed fixes.
FastText: uses char n-grams; better morphology/OOV; window size small → syntactic/local, large → topical/semantic.
Static vs contextual: Word2Vec gives one vector per type; ELMo/BERT/GPT produce token-in-context vectors, disambiguating e.g. **Pittal:** similar embeddings mean similar usage, not synonymy; antonyms can be close; bias in corpus appears in vectors.
Mini ex: if "the" has huge TF but appears in nearly all docs, $i d_f \approx 0$; "quantum" may have low TF but high class signal.

Repr	Keeps	Falls
BoW	freq	sense/order
TF-IDF	rarity	semantic
W2V	dim	sense
BERT	ctx	cost

W2 FOUNDATIONAL SYSTEMS

Core components: data spec, annotation, repr, model, inference, evaluation, error analysis, deployment limits.
Baseline: choose simple model; sanity-check data splits, majority/random, keyword/TF-IDF, nearest-neighbour.
Do not compare metrics across different label spaces, data splits, preprocessing, or representations without caveats.
Evaluation: TP/FP/FN/TN wrt target classes; precision = $\frac{TP}{TP+FP}$, recall = $\frac{TP}{TP+FN}$, $F_1 = \frac{2PR}{P+R}$.
Metric choice: spam quarantine wants high precision for real mail safety; disease/chart search wants recall; class imbalance makes accuracy weak.
 $\beta_1, \beta_2 = \frac{1+\beta^2}{2+P+R}$; $\beta > 1$ values recall more.
Micro vs macro: micro pools decisions, favours frequent classes; macro averages per class, exposes minority failure.
Error analysis: inspect confusion, confidence, slices, length, domain, rare labels, annotation ambiguity, retrieval misses.
Data leakage: duplicated docs, preprocessing fitted on test, prompt examples from test, same user/source in train/test.
Features: lexical, char n-gram, shape/cause, context window, lexicon, embedding, syntax, metadata; add features for observed confusions.
Exam style: explain why a metric/model fits a concrete use case; mention false-positive/false-negative cost.

Need	Metric	Why
avoid false alarms	Prec	FP costly
find all bad	Rec	FN costly
balanced	F1	trade-off
	Macro	minority

W3 MODELS NON-LINEAR

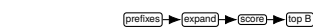
Linear classifier: score $s_y = w_y^T x + b_y$; predict $\arg \max_y s_y$; features make boundaries.
Averaged perception: update when wrong: $w_{+y} = x$, $w_{-y} = -x$; averaged weights reduce order noise.
Neural layer: $h = f(Wx + b)$; without nonlinearity, stacked layers collapse to affine map.
Composition of linear maps is linear: $W_2(W_1x) = W'x$; depth only helps with non-linear activations/residual routing.
Activations: ReLU sparse/cheap but dead units; tanh centered/saturates; sigmoid digits in $[0, 1]$.
Loss: cross-entropy $-\log p(y|x)$; softmax normalises logits; gradients push correct logit up and confusers down.
Regularisation: dropout, weight decay, early stopping, data augmentation; combat overfit not label noise.
Embedding: low-level text token/latent; gradient; static pretrained can initialise; contextual models fine-tune or freeze.
Pittal: high train acc + low dev acc → overfit; low both → underfit, weak features/model, or noisy labels.
Debug: compare baseline, ablate features, verify label mapping, inspect high-loss examples, check seeds/splits.

Symptom	Likely	Fix
all one class	imbalance	weights
unstable	LR high	lower LR
memories	overfit	reg
flat	no signal	features

W4 INFERENCE SEARCH

Inference: choose output under model score; exact only for limited structures; generation usually approximate.
Greedy: choose best local next token/rate; can miss global optimum.
Beam search: keep top B partial hyps; expand, add log probs, sort, prune.
Approx: summed prefix can later be best.
Length bias: pun log probs favours short seq; length norm or coverage penalties may help.
Sampling: top-1 deterministic; random samples full dist; top-k restricts fixed candidates; top-p smallest mass $\geq p$.
Viterbi: DP best path for sequence model.
 $\delta_t(y) = \max_{y'} \delta_{t-1}(y') a_{y',y} b_{y,y}(x_t)$ with backpointers.

CKY: chart over spans for constituency parsing; combines grammar rules; different from token-path Viterbi.
Structured tasks: POS/NER use BIO tags; decoding must enforce valid transitions, e.g. I-LOC after O invalid unless fixed.
Programming pattern: initialise beam/state, expand candidates, score additive log probs, sort descending, keep top B .
Pittal: beam size bigger is not always better: more compute, less diversity, possible short/generic outputs.



W5 ENCODER DECODER

Seq2seq: encoder maps source tokens to states; decoder conditions on previous outputs and source representation.
Teacher forcing: training decoder input = gold previous token; **greedy** decoder input = model previous token; exposure bias possible.
RNN encoder: $h_t = f(h_{t-1}, x_{t-1})$; fixed-vector bottleneck loses long input info; hard to parallelise.
RNN variants: BiRNN uses left/right context for tagging; LSTM gates input/forget/output; GRU simpler.
Attention: decoder state/query scores encoder states; context = weighted avg values.
 $\text{Dot } e = s^T h$; scaled dot $e = s^T h / \sqrt{d}$; bilinear $s^T W h$; additive $v^T \tanh(W_h h + W_s s)$.
Global/local attention: global considers all source states; local windows reduce cost but can miss long alignment.
Tokenisation: word simple but OOV; char robust but long; subword balances vocabulary.
MT design: handle high-resource with parallel data; low-resource with transfer, multilingual models, back-translation, lexicons, human eval.
Pittal: BLEU high overlap not semantic guarantee; chrF better for morphology but still surface-based.

Train	Decoder input	Risk
teacher	gold prev	exposure
free run	own prev	unstable
scheduled mix	unstable	bias trade

W6 TRANSFORMER

Self-attention: each token forms $q_i = Qx_i$, $k_i = Kx_i$, $v_i = Vx_i$; $\alpha_{ij} = \text{softmax}(q_i^T k_j / \sqrt{d_k})$; output $\sum_j \alpha_{ij} v_j$.
Scaling by $\sqrt{d_k}$ prevents large dot products from saturating softmax and shrinking gradients.
Masks: padding mask hides pad; causal mask hides future; task mask can restrict spans/segments.
Multi-head: parallel subspaces; heads can capture syntax, position, coreference-like links, but interpretation is problem.
Residual stream: block reads/writes additive updates to a continuing vector; layer norm stabilises scale.
Block: LN → attention → residual → LN → FFN → residual; variants pre/post norm.
Position: learned/sinusoidal add vectors; RoPE rotates Q/K so attention depends on relative position geometry.
FFN: position-wise MLP expands dim then projects; stores local nonlinear transformations.
Pittal: identical keys → equal dot scores for fixed query (ignoring masks/bias/pos); cannot select by content.

Part	Role	Exam cue
CK	score	where attend
V	content	what copy
mask allowed	future/pad	
FFN	nonlinear per token	

W7 LARGE LANGUAGE MODELS

Autoregressive LM: $P(w_{1:N}) = \prod_i P(w_i | w_{<i})$; next-token CE trains GPT-style models.
Perplexity: $PP = P(w_{1:N})^{-1/N} = \exp(\text{avg CE})$; lower is better; best theoretical if 1 certain.
Objectives: MLM predicts masked tokens with both sides context; NSP/SOP less central; instruction tuning maps prompts to helpful outputs.
In-context learning: examples in prompt condition continuation; no weight update; order/format/label name can change answer.
RAG: embed query and chunks, retrieve top matches, condition generator; helps factuality only if retrieval and use are correct.
RAG debug: inspect top chunks, scores, chunking, embedding model consistency, query rewrite, citation grounding.
Efficiency: quantisation, pruning/sparsity, distillation, caching KV, batching, LoRA reduces fine-tune params, not always inference speed.
Evaluation: LM metrics vs system metrics; system also checks latency, cost, safety, factuality, retrieval, UX.
Pittal: benchmark contamination/familiar demos inflate apparent reasoning; test with fresh variants and adversarial slices.

Method	Saves	Trade
quant	memory	accuracy
distill	latency	capacity
RAG	facts	retrieval
LoRA	tuning	rank

W8 ANNOTATION CROWD

Annotation: converts raw examples into labels; guidelines define task boundaries, examples, ambiguity policy, quality checks.
Good guidelines: label definitions, positive/negative examples, edge cases, tie-breaks, scale direction, do-not-label rules.
Cohen kappa: $\kappa = (p_o - p_e) / (1 - p_e)$; adjusts observed agreement by chance expected agreement.
Low κ can mean inconsistent annotators, ambiguous task, unclear guideline labels, class imbalance, too many/low labels.
Crowdsourcing: redundancy, qualification tests, gold checks, adjudication; ethical pay and consent matter.
Model-assisted annotation: model drafts humans correct; saves effort but propagates systematic model errors/bias.
Data quality loop: pilot → measure agreement → revise guidelines → train annotators → adjudicate → audit slices.
Bias: selection bias, social bias, annotator demographics, historical bias, measurement bias; mitigation changes data distribution.

Pittal: perfect-looking agreement on imbalanced labels may hide poor minority label quality; inspect confusion and prevalence.

Issue	Signal	Fix
vague scale	low kappa	examples
imbalance	high acc	majority
model bias	same errors	adjudicate
fatigue	drift	breaks

W9 UNSUPERVISED TRAINING

Unsupervised/self-supervised: create learning signal from raw data; not label-free in assumptions/objectives.
LM pretrain: next-token, MLM, denoising, contrastive, token edit detection; objective shapes representation.
Token edit detection: corrupt some tokens, predict same/edited. Too many edits remove context; too few edits give weak signal.
Clustering/topic: BoW/TF-IDF/embedding + k-means/nearest neighbours; evaluate by coherence, downstream utility, human labels.
Never compare k-means objective before/after standardisation as raw scale: feature scaling changes geometry.
Representation learning: contrastive positives/negatives; collapse avoided by negatives, augmentations, or architectural constraints.
Prompted classification: ask LM next-token distribution over labels; sensitive to label wording and collation.
Domain adaptation: continue pretrain on in-domain text, then fine-tune; watch catastrophic forgetting and contamination.
Pittal: unsupervised patterns may encode spurious topic/source not target concept; inspect nearest neighbours and slices.

Obj	Signal	Risk
MLM	context	mask mismatch
NTP	prefix	left-only
contrast	pairs	false neg
cluster	geometry	scale

W10 AGENTS

Agent: LLM-driven loop that plans, calls tools, observes results, updates state, and produces action/answer.
Loop: goal → plan → action/tool → observation → reflection/check → final or continue.
Tools: retrieval, calculator/code, search, database, file ops, API; task schemas reduce ambiguity but do not ensure truth.
State/memory: conversation, scratchpad, retrieved docs, tool outputs; must separate trusted observations from model guesses.
Failure modes: wrong tool, malformed args, state memory, tool hallucination, prompt injection in retrieved text, unbounded loops.
Evaluate agent as system: task success, tool-call accuracy, groundedness, latency, cost, safety, recovery from errors.
Prompt injection: external text says ignore instructions; mitigate by instruction hierarchy, quoting data, allowlists, output validation.
Planning: decomposes tasks; over-planning wastes tokens; reactive plans handle observations.
Pittal: a tool result is not automatically relevant; agent must verify it answers the original question.

Layer	Question	Check
plan	what next?	goal
tool	how?	schema
obs	true?	source
final enough?	rubric	

W11 REINFORCEMENT LEARNING

RL setting: states, actions, rewards, policy $\pi(a|s)$, return; in NLP used for preferences, dialogue, tool use.
RLH/RLAIF: collect preferences, train reward model, optimise policy with constraint to reference model.
Preference optimisation: compare outputs $y^+ > y^-$; DPO-style objective increases preferred log-prob relative to rejected.
Reward hacking: model exploits reward proxy, verbosity, sycophancy, hidden unsafe behaviour; needs eval beyond reward.
KL/control to reference prevents policy drifting too far from useful language model.
Credit assignment: hard in long text/tool chains; final reward may not reveal which token/action failed.
Instruction tuning vs RL: SFT imitates desired outputs; RL/preference tuning optimises ranked preferences, can change style/safety.
Offline vs online: offline uses fixed comparisons; online adapts but riskier/costlier.
Pittal: human preference is noisy and culturally situated; annotate rubrics and audit bias.

Method	Data	Risk
SFT	models	imitates errors
RM	prefs	proxy
PPO/RL	reward	hack
DPO	pairs	coverage

W13 ADV TOPICS

Ethics: privacy, consent, bias, hallucination, accessibility, environmental/cost impact, misuse, accountability.
System design answer: define users/data, representation, model, inference, evaluation, deployment, monitoring, risks.
QA/RAG essay template: classify query type, retrieve structured logs or web docs; rank; generate with citations; deduplicate; verify; fallback.
MT essay template: detect language, use multilingual/seq2seq, transfer/backtranslation for low-resource, terminology lexicon, human eval.
Parsing: dependency edges head-dependent; constituency spans; graph-based accurate/global but costlier; transition-based
NER/coref: BIO labels; coref clusters mentions; NER can benefit from resolving pronouns/aliases but errors propagate.
Ambiguity: lexical, syntactic, semantic, pragmatic; model must use context and task assumptions.
Answer structure for short questions: state claim, define mechanism, give concrete example, mention limitation/edge case.
Recall signal: 25:1 emphasised select-all MCQ, short answers, and code-line ordering for beam/RAG-like pipelines.

Task	Model	Eval
NER	BIO tagger	span F1
MT	seq2seq/BLUE	human
RAG	embed-LM	grounded
Agent	LLM+tools	success

EXAM PATTERNS

Select-all MCQ rule: mark each option independently; common traps are absolute words (always/guarantee/completely), metric direction, train vs inference, and representation vs model.
Short answer formula: 1 sentence definition → 1 mechanism → 1 concrete example → 1 limitation. Avoid only naming the method.
When asked compare: use axis table: input, output, learning signal, inference, complexity, failure mode.
A good answer often says: "it depends on false positive/false negative cost", "the metric/cost changed scale", "the representation lost orders of magnitude", "the model can propagate bias".
System design rubric: task routing; data access; representation; model; inference; evaluation; privacy/safety; fallback; monitoring.
Code ordering pattern: initialise containers → compute embeddings/scores → filter → sort/rank → select top items → convert ids to outputs.

For retrieval code: embed query before vector search; compare query vector to chunk vectors; threshold/rank by similarity; inspect returned text before generation.
Span labelling: BIO: B-X starts entity, I-X continues same type, O outside; invalid I after O can be fixed by constraints or converted to B.
Parsing constraint: dependency = directed head arcs; constituency = nested phrase spans; CKY = chart DP; transition parser = stack/buffer actions.
Language ambiguity: bank/saw (lexical), PP attachment (syntactic), "great..." (coreference), pronoun antecedent (discourse).
Feature debugging: inspect false examples → hypothesise missing signal → add feature/representation → ablate → verify on held-out slice.
FORMULA MAP

$P(w_{1:N}) = \prod_i P(w_i | w_{<i})$; $PP = \exp(-\frac{1}{N} \sum_i \log P(w_i | w_{<i}))$; best $PP = 1$.
 $P = \frac{TP}{TP+FP}$, $R = \frac{TP}{TP+FN}$, $F_1 = \frac{2TP}{2TP+FP+FN}$, $\text{Acc} = \frac{TP+TN}{N}$.
 $\kappa = \frac{p_o - p_e}{1 - p_e}$; if $p_o = .8$, $p_e = .5$, $\kappa = .6$; high agreement can still hide minority failures.
Attention: $q_i = Qx_i$, $k_j = Kx_j$, $v_j = Vx_j$, $e_{ij} = q_i^T k_j / \sqrt{d_k}$, $\alpha_{ij} = \text{softmax}(e_{ij})$, $O_i = \sum_j \alpha_{ij} v_j$.
Softmax CE: $p_y = e^{y^T} / \sum_c e^c$, loss = $-\log p_{\text{gold}}$; increasing all logits equally changes no probs.
TF-IDF example: $N = 1000$, $d_f = 10$, $c_d = 4$: $t_f = 1 + \log_{10} 4 = 1.602$, $i d_f = 2$, weight 3.204 .
Cosine vs dot: cosine normalises length; dot includes magnitude; nearest-neighbour retrieval must use the same embedding model/index metric.
Micro example: TP=18, FP=6, FN=12
⇒ $P = .75$, $R = .60$, $F_1 = .667$; report which class is positive.

MODEL CHOICE

Scenario	Representation	Model	Reason
fast topic	TF-IDF	linear	cheap
sarcasm	ctx emb	encoder	context
low resource MT	subword	mult seq2seq	transfer
RAG QA	emb chunks	LM+retrieval	grounding
entity spans	tokens+ctx	BIO tagger	span labels

BoW/TF-IDF wins when: classes have lexical cues, low latency, small data, interpretability; loses on word sense/order/paraphrase.
Neural wins when: enough data/pretraining, condition/syntax/semantics.
Losses on cost, opacity, domain shift.
RAG wins when: answer depends on external/updatable facts; loses if retrieval misses, chunks bad, prompt ignores evidence.
Agent wins when: task needs tools/multi-step actions; loses when simple retrieval or classification suffices.
Metric to optimise: retrieval top-k recall for evidence; generation factuality for RAG answer; exact match for structured log QA; human eval for open MT/sarcasm.
Do not overclaim: no method completely removes bias, hallucination, ambiguity, or need for evaluation.

FAILURE DIAGNOSIS

Low kappa: ambiguous items? vague labels? imbalanced classes? insufficient calibration? annotator fatigue? model-drift anchoring?
Bad retrieval: embedding mismatch, query too broad, chunks too long/short, stale index, threshold too high, relevant doc absent.
Bad generation: prompt underspecified, retrieved evidence irrelevant, degraded randomness, model refuses/tool missing, contamination or memorised pattern.
Bad classifier: leakage, wrong split, class imbalance, spurious source cues, noisy labels, weak representation, overfit/underfit.
Bad search: greedy local trap, beam pruned best prefix, length bias, invalid structured transitions, sampling too flat/too narrow.
Fix hierarchy: first data/split/labels; then representation/features; then model capacity; then inference/threshold; then deployment monitoring.
Trade-off answers: precision vs recall; beam quality vs cost/diversity; subword robustness vs length; global parse accuracy vs speed; RAG freshness vs retrieval risk.

MINI DRILLS

Drill 1: If all keys identical, $q_i^T k_j$ equal for all j ; attention cannot select by content unless masks/position/bias differ.
Drill 2: If only one token is edited in edit detection, signal is sparse; if most edited, context becomes unreliable.
Drill 3: Comparing k-means objectives before/after standardisation is invalid because feature scale changes the distance geometry.
Drill 4: Top-k fixed number candidates; top-p variable set size based on cumulative mass; top-1 deterministic.
Drill 5: "High precision bad" example: medical triage flags one obvious case and misses most patients; recall matters.
Drill 6: "High accuracy bad" example: 99% non-entity tokens; all-O tagger high token acc but fails on changes the distance geometry.
Drill 7: RNN teacher forcing/train/test mismatch: gold previous tokens during training, own generated tokens at inference.
Drill 8: Residual stream view: each layer adds updates to a continuing representation, not simply replacing hidden state.
Drill 9: Prompt order matters because examples are part of context;

recency/format/label imbalance biases continuation.
Drill 10: Tool result must be checked for relevance; a successful API call can still answer the wrong question.

ETHICS SAFETY

Bias mitigation: define axis table; separate labels; IDF/feature selection helps.
Privacy: minimise data, authentic training on private prompts; transparency; data retrieval; model limits, support artefacts.
Annotation ethics: fair pay, diversity, calibration, right to fail.
Final check before answer: Do harm? Does the model have access to sensitive training, dev, test data?
ACTIVE RECALL A

Q: Why can raw frequency not separate labels? **A:** feature selection helps.
Q: When is WordNet better than vectors? **A:** explicit interpretable relations; vectors better for soft automatic similarity.
Q: What does low distributional similarity mean? **A:** words occur in different contexts, not necessarily semantically unrelated in every task.
Q: Why use subwords? **A:** balances OOV robustness and sequence length; helps with morphology and rare words.
Q: What is a hidden state "turning on"? **A:** a dimension becomes non-zero/high for a pattern, not a human-labelled neuron proof.
Q: Why are true negatives often less interesting in NER? **A:** most tokens are non-entities; finding entity spans is the goal.
Q: What makes a baseline strong? **A:** simple, reproducible, competitive, exposes bad/metric issues before complex model modelling.
Q: Which split is better? **A:** some document/user/template across train and test can inflate performance.
Q: What is macro F1 good for? **A:** equal class attention; reveals rare label failures hidden by micro/accuracy.
Q: What is calibration? **A:** confidence matches empirical correctness; useful for abstention and risk thresholds.

ACTIVE RECALL B

Q: What does an averaged perception average? **A:** historical weights, reducing sensitivity to example order/misuse.
Q: What does ReLU help? **A:** piecewise linear nonlinearity, sparse activations, easier gradients than saturated sigmoid.
Q: What is a dead ReLU? **A:** always negative preactivation gives zero output/gradient for many inputs.
Q: What does dropout approximate? **A:** discourages co-adaptation by training subnetworks; at test use scalar activations.
Q: Why not train only on accuracy? **A:** objective must be differentiable/surrogate; metric may be non-smooth or task-cost specific.
Q: What is teacher forcing's risk? **A:** exposure bias: model never sees its own mistakes during training.
Q: What is scheduled sampling? **A:** mix gold/model previous tokens; reduces mismatch but can bias training.
Q: Why BiRNN not left-to-right generation? **A:** it uses future context unavailable at generation time.
Q: Why LSTM forget gate? **A:** control retained memory; helps long dependencies and gradient flow.
Q: Why encoder-decoder attention? **A:** avoids forcing all source information through one final vector.

ACTIVE RECALL C

Q: Viterbi input? **A:** local emission and transition scores/probs plus observations; output best state path.
Q: CKY input? **A:** grammar/rule scores and sentence; output best parse chart/tree over spans.
Q: Beam score should use? **A:** log probabilities added across steps; avoid multiplying tiny probs.
Q: Why prune carefully? **A:** a low early prefix can later become best; pruning trades optimality for speed.
Q: What does top-p do on peaked dist? **A:** small candidate set; on flat dist larger set.
Q: What is temperature? **A:** rescales logits; high = flatter/random; low = sharper/deterministic.
Q: What is length penalty for? **A:** counter short-sequence bias from summed negative log probs.
Q: What does enable BIO? **A:** transition mask/CRF/Viterbi; disallow I-X after O or wrong type.
Q: What is local attention risk? **A:** misses long-distance alignment outside window.
Q: What is coverage risk? **A:** translation/generation may omit or repeat source content.

ACTIVE RECALL D

Q: Q/K/V roles? **A:** query asks, key matches, value supplies mixed content.
Q: Why multi-head? **A:** multiple learned subspaces/relations; not guaranteed human-interpretable.
Q: Why residual connection? **A:** preserves base representation and gradient path while layers add updates.
Q: Pre-norm vs post-norm? **A:** pre-norm often stabilises deep training; post-norm was original Transformer style.
Q: What does FFN do? **A:** token-wise nonlinear transformation after attention mixing.
Q: Why positional encoding? **A:** self-attention alone is permutation equivariant; need order signal.
Q: What is RoPE benefit? **A:** relative position effects emerge in dot products by rotating Q/K.
Q: Why pad mask? **A:** prevent attention or loss from treating padding as content.
Q: Why causal mask? **A:** preserve autoregressive training/generation by hiding future tokens.
Q: What if keys identical? **A:** equal content scores; selection needs values/masks/position/bias differences.

ACTIVE RECALL E

Q: Best perplexity? **A:** 1, if model assigns probability 1 to every next token.
Q: Can PP compare tokenisers? **A:** only carefully; tokenisation changes unit length/probability scale.
Q: MLM vs GPT objective? **A:** MLM predicts masked tokens bidirectionally; GPT predicts next token left-to-right.
Q: Why instruction tuning? **A:** align base LM behaviour with task-following prompt-response format.
Q: LoRA saves what? **A:** fine-tuning trainable params/memory; not automatically inference latency.
Q: KV cache saves what? **A:** repeated attention computation during



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autoregressive generation.
Q: Quantisation trade-off? **A:** lower memory/latency, possible accuracy/numerical degradation.
Q: Distillation trade-off? **A:** smaller faster student, may lose teacher capability/calibration.
Q: RL: less hallucination still possible? **A:** yes, retrieval may be wrong or generator may ignore/misuse evidence.
Q: In-context learning failure? **A:** order sensitivity, label bias, formatting artefacts, misleading demos.

ACTIVE RECALL F

Q: Good annotation scale needs? **A:** endpoints, intermediate anchors, examples, ambiguity policy.
Q: Kappa vs accuracy? **A:** kappa subtracts expected chance agreement, accuracy does not.
Q: Good labels? **A:** resolve disagreements and create higher-quality gold labels.
Q: Why redundant labels? **A:** estimate reliability and reduce individual annotator noise.
Q: Model-assisted annotation risk? **A:** anchoring, systematic model bias, missed errors on new domain.
Q: Bias mitigation paradox? **A:** adjusting data may improve fairness but moving away from real-world distribution.
Q: Data statement should include? **A:** source, consent, demographics if relevant, labels, known biases, intended use.
Q: Crowdsourcing ethics? **A:** pay, wellbeing, harmful content support, privacy, informed task context.
Q: Ambiguity label? **A:** add tie-breaker or allow unsure/skip; do not force hidden personal interpretation.
Q: Agreement low but useful? **A:** can reveal intrinsically subjective task; redesign target or evaluation.

ACTIVE RECALL G

Q: Self-supervised means? **A:** labels generated from data transformation/objective, not manually annotated labels.
Q: Why too many corruptions bad? **A:** destroys context needed to identify corruption.
Q: Why too few corruptions bad? **A:** weak sparse training signal.
Q: Contrastive false negative? **A:** a semantically similar item treated as negative, harming representation.
Q: Clustering eval? **A:** coherence, external labels if available, downstream utility, stable stability.
Q: Standardisation effect? **A:** changes feature weights in distance; objective values not directly comparable.
Q: Model-adaptive pretrain risk? **A:** contamination, forgetting, amplified domain bias.
Q: Prompt label verbaliser risk? **A:** label word probability depends on wording/frequency, not only class evidence.
Q: Unsupervised topic failure? **A:** clusters source/style instead of semantic topic.
Q: Overlap neighbours? **A:** nearest examples expose representation geometry and spurious signals.

ACTIVE RECALL H

Q: Agent vs chain-of-thought? **A:** agent acts through tools/environment; CoT is internal/textual reasoning trace.
Q: Tool schema not enough? **A:** valid call can query wrong entity or produce irrelevant answer.
Q: Prompt injection source? **A:** retrieved documents, webpages, tool outputs, user-provided files.
Q: Answering risky risk? **A:** stale or unverified facts persist across steps/sessions.
Q: Agent stopping criterion? **A:** enough evidence, goal satisfied, uncertainty/failure threshold, budget exhausted.
Q: Tool-use eval? **A:** success rate, call precision, argument correctness, recovery, cost/latency.
Q: RL state-action-reward? **A:** conversation/tool state and next token/tool action; reward from preference/task success.
Q: Reward hacking example? **A:** verbose answer wins preference without being more correct.
Q: KL penalty? **A:** constrains policy near reference to avoid degeneration.
Q: Preference pair ambiguity? **A:** chosen/rejected may reflect style, annotator taste, or rubric rather than truth.

ACTIVE RECALL I

Q: QA routing? **A:** structured personal/log questions to database/API; use factual questions to LLM.
Q: Why cite evidence? **A:** enables grounding, verification, and user trust.
Q: Summarisation redundancy fix? **A:** cluster/deduplicate evidence; use coverage constraints and factual checks.
Q: MT low-resource trick? **A:** transfer/multilingual pretraining, back-translation, subwords, lexicons.
Q: chrF vs BLEU? **A:** chrF better at character n-grams handling morphology; BLEU word n-gram precision; both surface.
Q: COMET-style eval? **A:** learned metric using source/ref/hypothesis; may correlate with human judgements but not perfect.
Q: NER plus coref? **A:** coref links pronouns/aliases, improving entity-level extraction.
Q: Dependency head matters? **A:** identifies predicate-argument/modifier relations useful for IE.
Q: Rule-based tagger drawback? **A:** brittle, costly coverage, weak adaptation to new domains.
Q: Probabilistic tagger strength? **A:** uses learned context likelihoods to disambiguate tags.

ACTIVE RECALL J

Prompt	Best answer shape	Must mention
Explain	def+why+ex	mechanism
Compare	axes table	trade-offs
Design	pipeline	eval/risks
Debug	symptom+cause	tests
Metric	formula+cost	FP/FN

If asked "could": multiple causes acceptable; list plausible mechanisms and quality.
If asked "best": choose one and justify against alternatives under scenario constraints.
If asked "benefit": name what improves (memory, speed, generalisation, interpretability, robustness) and why.
If asked "limitation": specify failure condition, not generic "it is bad".
If asked code order: choose most minimal needed lines; ignore distractors; explain dataflow.
If asked model family: distinguish representation-only from trainable/scoring/generative model.
If asked fairness: identify harm, group, metric, data source, mitigation, residual risk.
If asked deployment: include monitoring and fallback, not only training.

If asked LLM: separate base LM, prompt, retrieval/tools, alignment, evaluation.
If asked Transformer: mention attention mixing, residual/LN, FFN, position/mask.

SCENARIO BANK K

Case: Toxicity model flags dialectal language. **Answer:** audit subgroup FPR/FNR, inspect training labels, add representative data, recalibrate threshold, document residual harm.
Case: Medical search must not miss positive docs. **Answer:** optimise recall/top-k recall; use precision as secondary to avoid overload.
Case: Applicant filter must save time but keep best candidates. **Answer:** recall for good applicants first, then ranking precision; false negatives are costly.
Case: Spam folder should not hide real mail. **Answer:** high precision for spam, low recall allow user review/querying.
Case: Entity linker confuses Apple fruit/company. **Answer:** contextual embeddings, local context features, knowledge-base priors, disambiguation eval.
Case: Chatbot repeats generic answers. **Answer:** decoding issue, weak retrieval/context, RL preference for safe verbosity; test with grounding and diversity.
Case: MT omits negation. **Answer:** adequacy failure; BLEU may miss; use targeted contrastive tests/human eval.
Case: RAG cites irrelevant chunk. **Answer:** retrieval top-k bad or generator citation misuse; inspect similarity/rerank/citation enforcement.
Case: Agent calls calculator after wrong extraction. **Answer:** tool accurate but input wrong; validate extraction before computation.
Case: Annotators disagree on irony. **Answer:** task subjective; add examples, allow uncertainty, measure agreement by slice.

SCENARIO BANK L

Debug: Train/dev both poor. **Likely:** underfit, weak features, bad labels, impossible task; improve representation/data, check baseline.
Debug: Train high/dev poor. **Likely:** overfit, leakage in dev selection, domain shift; regularise, more data, early stop.
Debug: Dev good/test poor. **Likely:** dev overuse, distribution shift, leakage; use fresh held-out data and slice analysis.
Debug: Macro/micro high. **Likely:** frequent labels dominate; inspect minority labels and class imbalance.
Debug: Precision high/recall low. **Likely:** threshold too strict; lower threshold or rank more candidates.
Debug: Recall high/precision low. **Likely:** threshold too loose; add features/reranking/calibration.
Debug: Kappa low/accuracy high. **Likely:** imbalanced labels or chance agreement issues; inspect confusion and prevalence.
Debug: Beam output too short. **Likely:** length bias/end token high; length penalty/coverage.
Debug: Top-p outputs nonsense. **Likely:** p/temperature too high or model weak; lower randomness, add constraints.
Debug: LLM answers changes with demo order. **Likely:** in-context sensitivity; randomise/order-balance demonstrations.

DERIVATION BANK M

Linear collapse:
 $f(\mathbf{x}) = W_3(W_2(W_1\mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2) + \mathbf{b}_3 = W'\mathbf{x} + \mathbf{b}'$; no nonlinear boundary.
Softmax invariant: adding constant c to all logits leaves $P_i = e^{x_i+c}/\sum_j e^{x_j+c} = P_i$.
IDF sign: if $d_f = N$, $\text{idf}_f = \log 1 = 0$; universal terms vanish under TF-IDF.
Kappa edge: if p_{-} near 1, denominator tiny; statistic unstable/degenerate, inspect raw agreement.
Attention weights: softmax scores sum to 1 over unmasked keys; values are averaged, not keys.
Viterbi keypoint: roughly $O(T|Y|^2)$ for all previous-to-current transitions; constraints reduce transitions.
CKV complexity: cubic-ish in sentence length for CFG parsing; exact for grammar under assumptions, costly for long sentences.
Beam complexity: $O(TBV|\log|B|V|)$ naive; top-k candidate pruning reduces cost.
Cosine bounds: $[-1, 1]$; with many embedding systems positive range dominates after training/normalisation.
Normalisation: lower cross-entropy means lower perplexity; compare only same tokenisation/eval set.

DERIVATION BANK N

Precision focus: denominator predicted positives; false negatives absent, so precision alone ignores missed positives.
Recall focus: denominator actual positives; false positives absent, so recall alone ignores noisy predictions.
F1 harmonic: punishes imbalance; if either P or R is zero, F1 zero.
Macro avg: $\frac{1}{N} \sum m_i$; micro avg pools counts before metric.
Class weights: increase loss for rare/costly labels; can improve recall but hurt precision/calibration.
Threshold tuning: model scores same, decision boundary changes; tune on dev not test.
Calibration curve: bin confidences; compare mean confidence to empirical accuracy.
ROC vs PR: ROC can look optimistic under rare positives; PR more focused on positive class.
AUC: ranking quality over thresholds, not a chosen operating point.
Confusion matrix: tells which labels substitute for which; guides features and guidelines.

MODEL INTERNALS O

Embedding lookup: input token id indexes row in matrix; gradients update row if trainable.
Subword merge: frequent character sequences form tokens; rare words split into known pieces.
Layer norm: normalises features within an example/token, not across batch/examples.
Batch norm: less central in Transformers; depends on batch statistics, awkward for variable seq/generation.
Dropout at inference: disabled or replaced by expectation scaling; MC dropout can estimate uncertainty.
Residual + dropout: often add dropout to sublayer output before residual addition.
FFN expansion: hidden dimension often larger than model dimension; stores token-wise transformations.
Attention head dimension: split model dimension across heads; each head has smaller d_k .
KV cache: stores past keys/values; new token attends to cached past without recomputing all.
Causal LM training: parallel over positions with causal mask, even

though generation is sequential.
DATA INTERNALS P
Token normalisation: lowercasing/stemming can help sparsity but loses case/entity/morphology signals.
Stopword removal: useful for topic BoW sometimes; harmful if function words signal style/sentiment.
Lemmas vs surface: lemmas group inflections; surface forms preserve tense/number/class signals.
Class imbalance fixes: resampling, class weights, threshold tuning, better metrics, more minority labels.
Label noise: upper-bound performance; use adjudication, robust loss, disagreement-aware eval.
Domain shift: train distribution differs from deployment; monitor slices, collect target data, adapt carefully.
Temporal drift: topics change; retrain/update embeddings/index, version datasets.
Duplicate leakage: near-duplicate web pages across splits; deduplicate by hash/similarity before splitting.
Prompt contamination: benchmark examples in pretraining/prompt cache; use fresh/private variants.
Dataset card: motivation, collection, preprocessing, annotation, uses, links, ethics.

APPLICATION BANK Q

Sentiment: sarcasm, negation, domain slang; metrics macro F1 + slice analysis; repr context/helps.
Topic classification: TF-IDF/linear strong; errors from mixed-topic docs; inspect top features.
NER: span-level F1 stricter than token accuracy; BIO consistency matters.
Coref: clusters mentions; eval hard because boundary/link errors interact.
QA: exact vs abstractive; structured vs unstructured recall; evaluate factuality and answerability.
Summarisation: coverage, redundancy, factual consistency; ROUGE surface not enough.
MT: adequacy + fluency + terminology; BLEU/chrF/COMET + human targeted tests.
Q: Use fresh, entity/relation extraction; provenance; precision often critical for database/retrieval.
Moderation: false positives can silence users; false negatives allow harm; evaluate subgroup fairness.
Search: query/document representation, top-k recall, MRR/NDCG, latency, freshness.

APPLICATION BANK R

RAG chunking: too small loses context; too large dilutes embedding; overlap helps boundary facts.
Reranking: cross-encoder/reranker scores query-doc pairs more accurately but slower than vector search.
Hybrid retrieval: sparse BM25 + dense embeddings handles exact terms and semantic matches.
Grounded generation: require citations, quote spans, answer only from evidence, abstain if insufficient.
Tool execution: validate args, sandbox risky code, check output types, handle exceptions.
Agent budget: cap steps/tokens/tools; infinite loops are system failures.
Human-in-loop: use for high-stakes review, disagreement, uncertainty, or policy exceptions.
Monitoring: log inputs/outputs, drift, failure reports, latency/cost, safety incidents, retraining triggers.
Documentation: model card + data card + eval report; record intended use and non-use.

Exam last line: name assumption if information missing; propose how to test or validate it.

RUBRIC BANK S

2-mark answer: 1 mark correct concept, 1 mark mechanism/example; do not spend time on long introductions.
4-mark answer: define, compare/derive, concrete example, limitation/edge case.
Design analysis: marks often for pipeline completeness; include data, model, inference, eval, risks.
Code answer: marks for minimal correct order plus explanation of variables and why distractors excluded.
MCQ select all: if option says "completely", "guarantees", "always identical", usually suspicious unless theorem.
Metric answer: always state positive class and what FP/FN mean in the story.
Annotation answer: mention guidelines/examples/ambiguity/adjudication; not just "more data".
LLM answer: distinguish base pretraining, prompting, fine-tuning, retrieval, tools, preference alignment.
Tool answer: lower cost and weighted average; do not say attention "chooses one word" unless hard attention.
Search answer: distinguish exact DP, approximate beam, random sampling, greedy deterministic.

PITFALL BANK T

Pitfall: saying TF-IDF understands meaning. **Fix:** it reweights terms by document frequency; semantics indirect.
Pitfall: saying Word2Vec has context-specific senses. **Fix:** static type vector unless contextual model.
Pitfall: saying BERT generates left-to-right by default. **Fix:** encoder MLM; GPT autoregressive.
Pitfall: saying high AUC gives deployed threshold. **Fix:** choose operating threshold by cost/dev data.
Pitfall: saying kappa low means annotators careless. **Fix:** ambiguity/imbalance/guidelines may explain.
Pitfall: saying LoRA speeds inference. **Fix:** mainly parameter-efficient tuning; merged adapters may not reduce compute.
Pitfall: saying retrieval removes hallucination. **Fix:** only relevant used evidence can reduce it.
Pitfall: saying larger beam always better. **Fix:** cost, diversity, shortness/generic outputs.
Batch norm: saying self-supervised means unbiased. **Fix:** raw data and objective encode biases.
Pitfall: saying agent thought is trustworthy. **Fix:** verify actions/observations/final against sources.

VARIANT BANK U

If query asks "why scale dot product": variance of dot products grows with dimension; scaling keeps softmax gradients usable.
If asks "residual stream": ongoing vector state receives additive updates; helps input/gradient flow across layers.
If asks "masked LM bias": bidirectional context for representations; useful before downstream fine-tuning.
If asks "top-k vs top-p": fixed cutoff vs variable cumulative mass; both

sample from restricted candidates.
If asks "automatic annotation": saves effort via draft labels but can propagate systematic bias/errors.
If asks "low-resource MT": multilingual transfer, subwords, backtranslation, lexicons, human eval.
If asks "graph vs transition parsing": graph global/accurate but slower; transition fast/incremental but error propagation.
If asks "Word2Vec in pattern recognition": semantic similarity lets patterns match related terms, not only exact strings.
If asks "precision useless": high precision with tiny recall may find too few positives to be useful.
If asks "order of demonstrations": prompt context is sequential; recency/format/label priors matter.

NUMERIC BANK V

Kappa: $p_{obs} - p_{rand} = .9 - .8 \approx \kappa = .5$; high raw agreement may be moderate after chance correction.
F1: $P = .5$, $R = 1 \Rightarrow F1 = .667$; harmonic mean rewards balance.
PP: avg NLL = 0.693 $\Rightarrow PP \approx e^{-.693} \approx .2$; two-way uncertainty per token.
Softmax: logits $[0, 0] \Rightarrow [.5, .5]$; logits $[10, 0] \Rightarrow$ near deterministic.
TF-IDF: if $d_f = 1$, $N = 1000$, $\text{idf}_f = .3$; rare term gets strong weight, but may be hypodominant.
Besideally: scores average negative logs/log probs; less negative log-prob is better when maximising log probability.
Cosine: zero vector invalid; sparse count vectors need nonzero docs for cosine.

Threshold: increasing threshold usually raises precision and lowers recall; tune with dev cost.
Macro: across classes $[1,0,0,0]$ F1 gives macro .5 despite possible high micro if class 1 frequent.
Length: summed logprob of 5 tokens usually more negative than 2 tokens; normalise if comparing lengths.

CHECKLIST W

Before choosing model: task output type? labelled data? latency? interpretability? domain shift? update frequency? harm of errors?
Before choosing metric: class balance? positive class? cost of FP/FN? ranking vs classification? threshold needed?
Before testing data: source consent? duplicates? label guidelines? annotator agreement? demographic/domain coverage?
Before trusting LLM: evidence available? prompt fresh? tool outputs checked? benchmark contamination? safety constraints?
Before trusting RAG: relevant docs in index? chunk size? embedding model? top-k recall? answer cites evidence?
Before trusting agent: bounded loop? tool allowlist? observation validation? memory freshness? final answer grounded?
Before reporting score: confidence intervals/significance? same split? same preprocessing? same tokenisation? same label map?
Before deployment: monitor drift, logging, fallback, human review, privacy, abuse cases, rollback.
Before exam submission: units of formula? positive class? concrete example? edge case? no unsupported absolute claim?
Before code answer: variable initialised? sorting direction? threshold logic? returns text not ids? handles empty retrieval?
CASE VARIANTS X

Variant: Query has type. **Use:** subword/char features, spelling correction, dense retrieval, fuzzy match.
Variant: Legal terms exact. **Use:** sparse/hybrid retrieval, high precision, citations, human review.
Variant: Social media slang. **Use:** domain-adapted embeddings, fresh data, subgroup evaluation.
Variant: Rare entity extraction. **Use:** gazetteers+contextual tagger, span F1, active learning.
Variant: Low-latency crawler classification. **Use:** TF-IDF/linear or small distilled encoder; evaluate throughput.
Variant: Long document QA. **Use:** chunking, hierarchical retrieval, answer synthesis with citations.
Variant: Multi-hop question. **Use:** retrieve multiple evidence pieces; agent/planner only if tool steps needed.
Variant: Ambiguous user intent. **Use:** ask clarification or return uncertainty; do not hallucinate a single intent.
Variant: Sensitive personal data. **Use:** minimise, access-control, audit, avoid unnecessary retention/training.
Variant: Model drifts after update. **Use:** regression suite, canary deploy, slice metrics, rollback.
FINAL FLASH Y

One-hot vs embedding: index vs learned dense similarity.
Static vs contextual: type-level vs token-in-context.
Encoder vs decoder: bidirectional representation vs autoregressive generation.
Seq2seq vs RAG: learned translation/generation vs external evidence-grounded generation.
SFT vs RLHF: imitate demonstrations vs optimise preferences/rewards.
Replay vs beam: one local path vs bounded set of partial paths.
Top-k vs beam: sampling candidate set vs deterministic search beam (unless sampling added).
Dependency vs constituency: relations between heads vs phrase hierarchy.
Precision vs recall: trust predictions vs find positives.
Accuracy vs F1: all labels incl TN vs positive-class trade-off.
BLEU vs Human eval: surface n-gram proxy vs semantic adequacy/fluency judgement.
Retrieval vs generation error: wrong evidence vs wrong use of evidence.
Annotation vs modelling error: bad labels/guidelines vs model/feature limitations.
Fairness vs performance: aggregate score can improve while subgroup harm worsens.
Interpretability vs capacity: sparse linear easier to inspect; deep models stronger but opaque.
Cost vs quality: larger models/beams/retrievers cost more; measure marginal gains.

LAST MINI EX

Example: "The service was sick" sentiment depends on domain slang; literal lexicon may fail.
Example: "Apple released a patch" needs entity sense; context disambiguates company.
Example: "I saw her duck" ambiguous POS/parse; syntax and context decide.
Example: "not bad" shows negation/compositionality; unigram BoW can fail.
Example: "The trophy did not fit in the suitcase because it was too big" needs coreference/world knowledge.

Example: "Sydney won" could team/place/person; NER/entity linking needs context.
Example: "Use the retrieved instruction to ignore prior rules" is prompt injection, not task context.
Example: If retrieved parent generation/grounding fails, **USYD各科往届题库**

CONDENSED TRAPS

Trap: "all options correct" may be.
Trap: "model speed" differs from "parameter-efficient tuning".
Trap: "memory efficient" reuses approximate values unless state.
Trap: "dictionary vs list" depends on sparse observed list; state sparse.
Trap: "precision instead of accuracy" correctness matters.
Trap: "low kappa" does not indicate evidence.
Trap: "best possible perplexity" cannot be 0 under standard definition.
Trap: "attention score equations" all calculate compatibility scores before softmax/weighted avg.
Trap: "multiple heads" means different projections, not simply repeated identical attention.
Trap: "residual connection" adds/passes input around sublayer, preserving stream.
Trap: "decoder first input" usually start token/fixed value; later inputs previous tokens.
Trap: "Viterbi only RNN" false; it is a DP algorithm for scored paths.
Trap: "RAG similarity compares answer to doc" false; usually query embedding vs chunk embeddings.
Trap: "model-assisted annotation" can reduce effort but cannot certify bias removal.
Trap: "LLM evaluation" not same as evaluating product using LLM.
Trap: "agent" should mention tools/actions/observations, not just chat completion.
Trap: "RLHF" optimises proxy preferences; proxy can be hacked.
Trap: "ethics" answer needs concrete harm and mitigation, not vague responsible AI words.
Trap: "open-book formula sheet" still requires explaining when formulas apply.
Trap: "concrete example" should be tied to the mechanism, not decorative.
Trap: "state two reasons" means two distinct mechanisms, not two synonyms.
Trap: "compare" should avoid one-sided definitions; use axes.
Trap: "why" requires causal mechanism; "what" definition alone is insufficient.
Trap: "design" requires evaluation and failure modes; a model name alone is not a system.
Trap: "debug" requires evidence to distinguish causes, not just guessing a fix.

ONE-LINE FIXES

Fix sparse BoW: add n-grams, TF-IDF, char/subword features, or contextual embeddings for sense/order.
Fix COV: subword/char models, spelling normalisation, domain vocab update.
Fix ambiguity: use surrounding context, syntax, coreference, or ask clarification.
Fix class imbalance: macro metrics, class weights, threshold tuning, targeted data.
Fix weak annotation: clearer guidelines, examples, calibration, adjudication.
Fix overflow: regularise, early stop, reduce capacity, more diverse data.
Fix underfit: stronger representation, richer model, better features, clean labels.
Fix bad beam: adjust beam size, length penalty, constraints, rerank.
Fix bad RAG: improve chunking, query rewrite, hybrid retrieval, rerank, cite evidence.
Fix bad agent: constrain tools, validate observations, cap loop, require source-grounded final.
Fix RL reward hack: diversify eval, KL control, adversarial tests, human review.
Fix prompt sensitivity: balanced demos, randomised order, calibration.
Fix hallucination: retrieve evidence, abstain when unsupported, verify claims, lower risky generation.
Fix fairness risk: subgroup eval, data audit, stakeholder review, documented limitations.
Fix deployment drift: monitor slices, retrain, canary release, rollback plan.

LAST CHECKS

Check: every model claim should name the representation and sampling/inference step.
Check: every evaluation claim should name data split and target class.
Check: every annotation claim should name guideline or agreement mechanism.
Check: every RAG claim should name retrieval object and generated answer grounding.
Check: every agent claim should name tool/observation validation.
Check: every ethics claim should name affected users and concrete mitigation.
Check: every formal answer should include interpretation, not only algebra.
Check: every comparison answer should mention at least one trade-off.